

The GLM Procedure

Class Level Information		
Class	Levels	Values
region	3	Northern Italy Sardinia Southern Italy

Number of Observations Read	572
Number of Observations Used	572

The GLM Procedure

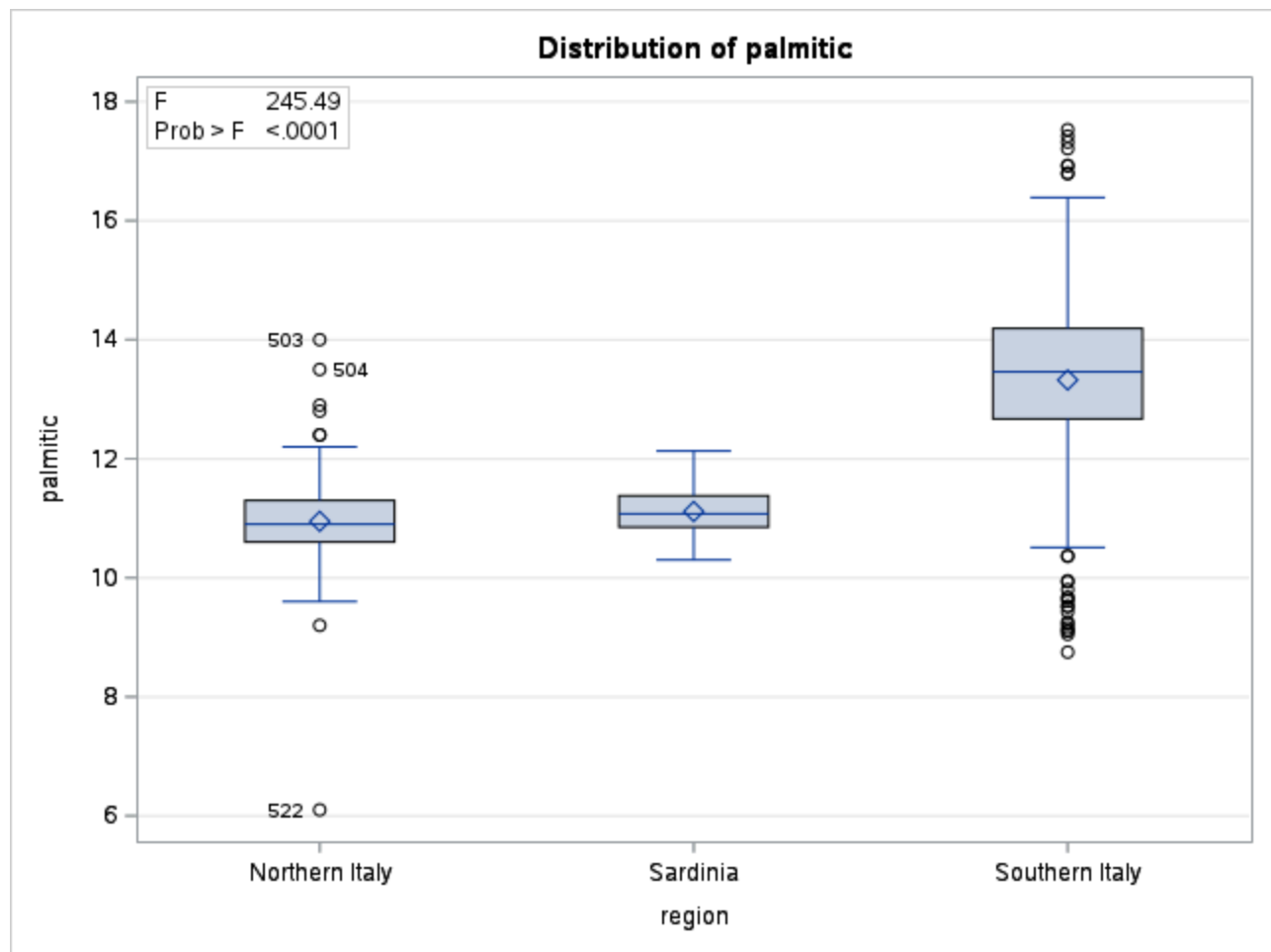
Dependent Variable: palmitic

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	751.753524	375.876762	245.49	<.0001
Error	569	871.219847	1.531142		
Corrected Total	571	1622.973371			

R-Square	Coeff Var	Root MSE	palmitic Mean
0.463195	10.04589	1.237393	12.31741

Source	DF	Type I SS	Mean Square	F Value	Pr > F
region	2	751.7535240	375.8767620	245.49	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
region	2	751.7535240	375.8767620	245.49	<.0001



The GLM Procedure

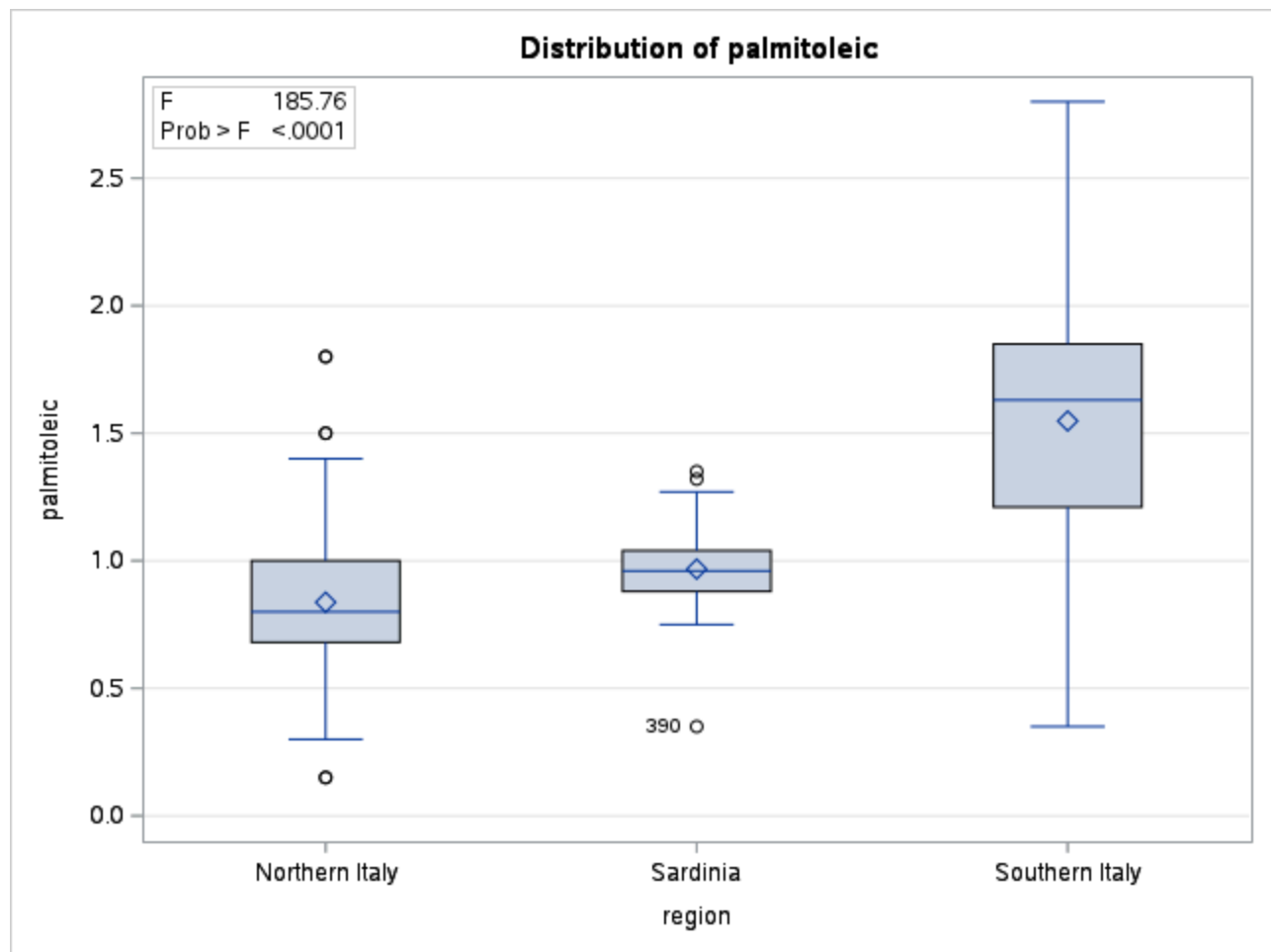
Dependent Variable: palmitoleic

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	62.1547557	31.0773778	185.76	<.0001
Error	569	95.1933345	0.1672994		
Corrected Total	571	157.3480902			

R-Square	Coeff Var	Root MSE	palmitoleic Mean
0.395014	32.43779	0.409022	1.260944

Source	DF	Type I SS	Mean Square	F Value	Pr > F
region	2	62.15475568	31.07737784	185.76	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
region	2	62.15475568	31.07737784	185.76	<.0001



The GLM Procedure

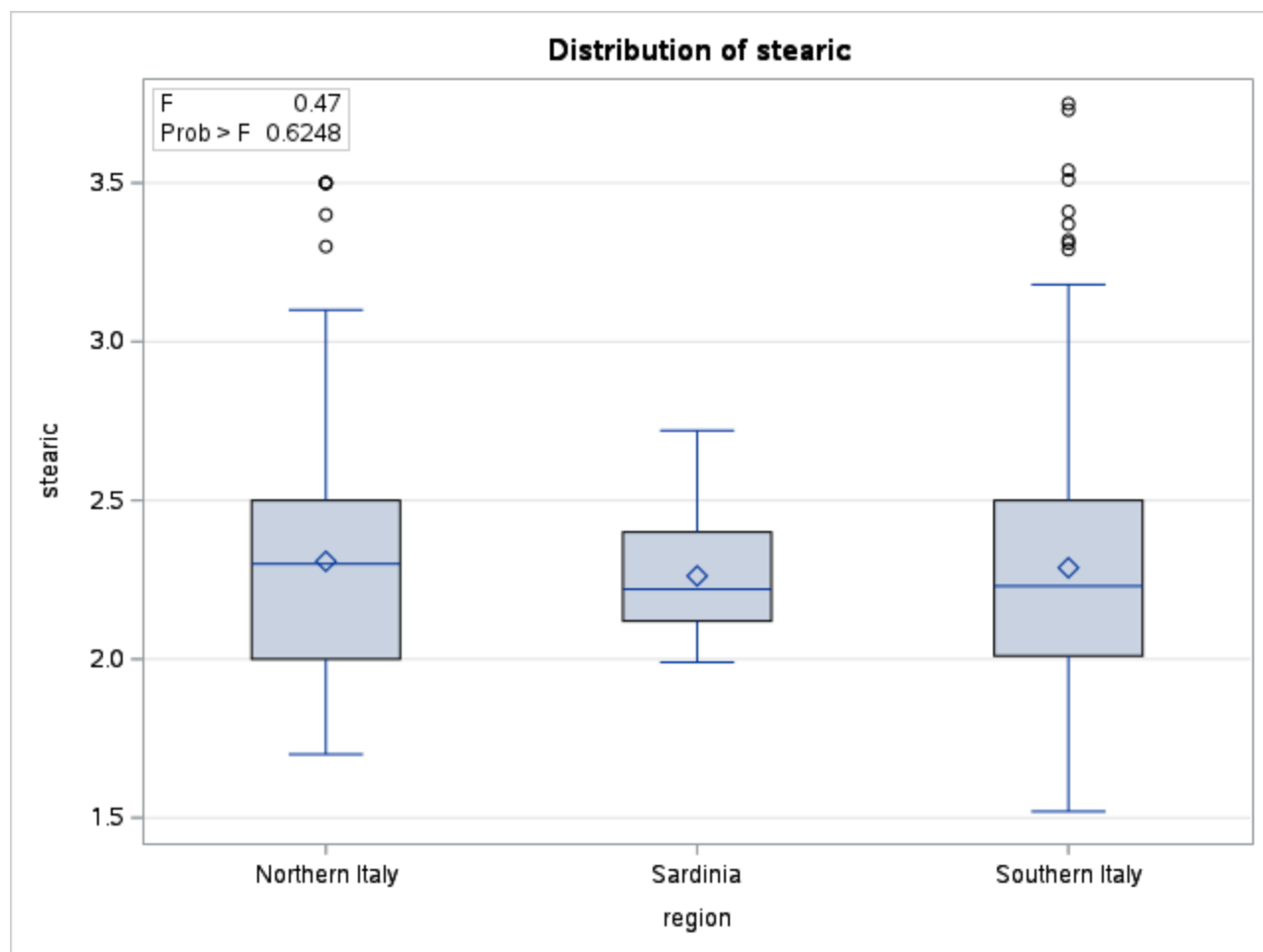
Dependent Variable: stearic

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	0.12733995	0.06366997	0.47	0.6248
Error	569	76.96852352	0.13526981		
Corrected Total	571	77.09586346			

R-Square	Coeff Var	Root MSE	stearic Mean
0.001652	16.07016	0.367790	2.288654

Source	DF	Type I SS	Mean Square	F Value	Pr > F
region	2	0.12733995	0.06366997	0.47	0.6248

Source	DF	Type III SS	Mean Square	F Value	Pr > F
region	2	0.12733995	0.06366997	0.47	0.6248



The GLM Procedure

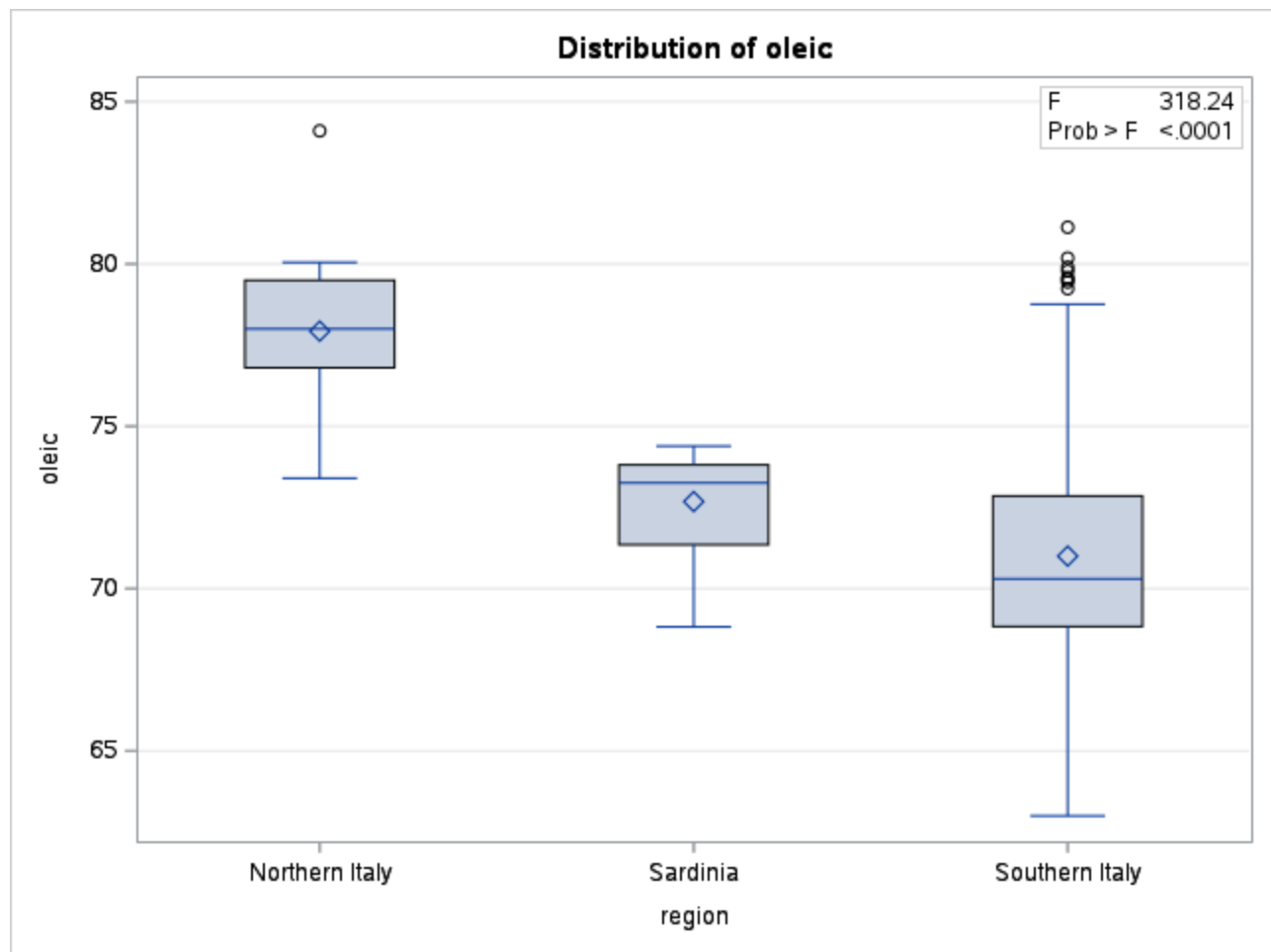
Dependent Variable: oleic

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	4964.836324	2482.418162	318.24	<.0001
Error	569	4438.502251	7.800531		
Corrected Total	571	9403.338575			

R-Square	Coeff Var	Root MSE	oleic Mean
0.527987	3.819802	2.792943	73.11748

Source	DF	Type I SS	Mean Square	F Value	Pr > F
region	2	4964.836324	2482.418162	318.24	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
region	2	4964.836324	2482.418162	318.24	<.0001



The GLM Procedure

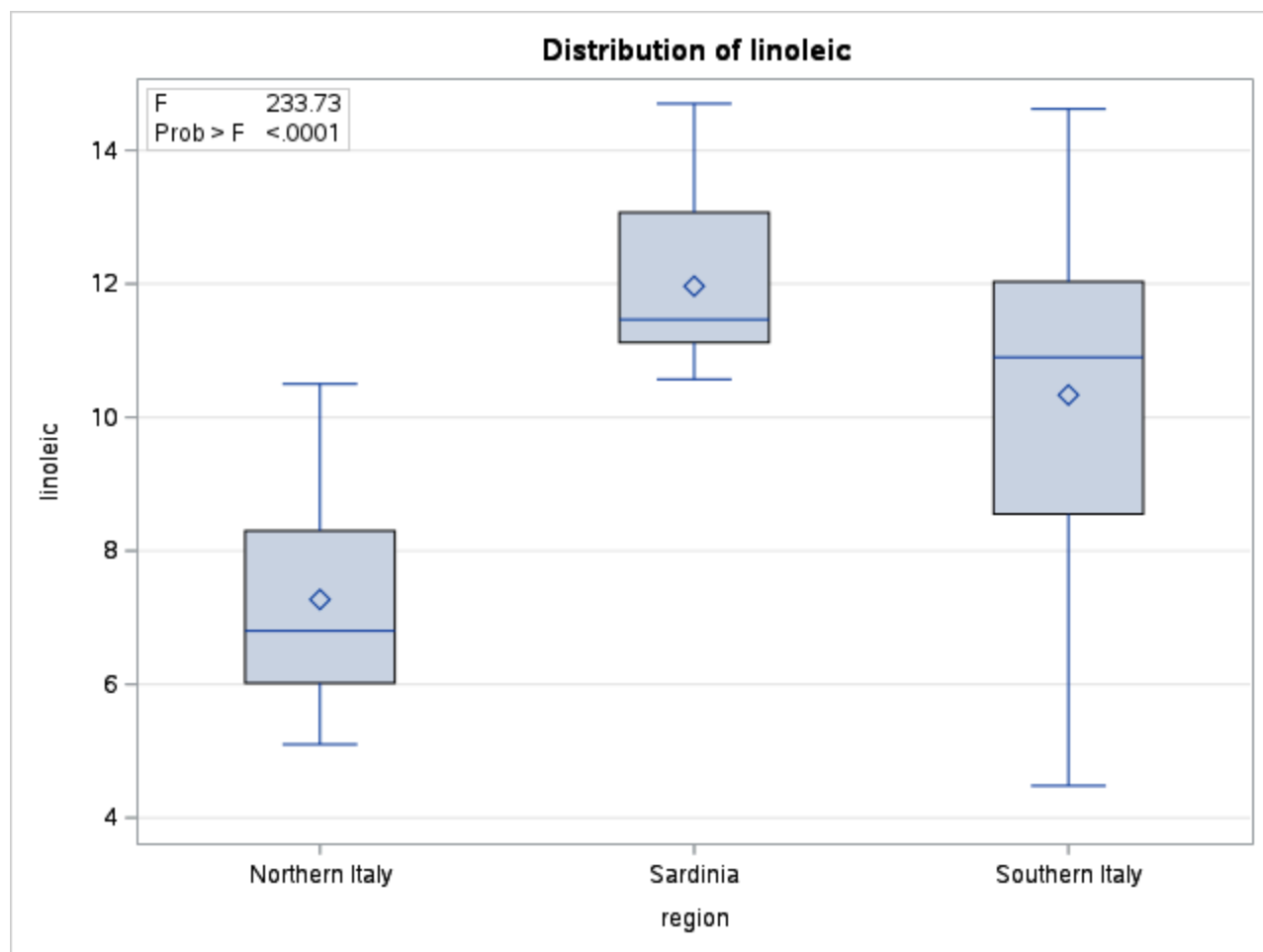
Dependent Variable: linoleic

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	1518.190256	759.095128	233.73	<.0001
Error	569	1847.938199	3.247695		
Corrected Total	571	3366.128455			

R-Square	Coeff Var	Root MSE	linoleic Mean
0.451020	18.37924	1.802136	9.805280

Source	DF	Type I SS	Mean Square	F Value	Pr > F
region	2	1518.190256	759.095128	233.73	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
region	2	1518.190256	759.095128	233.73	<.0001



The GLM Procedure

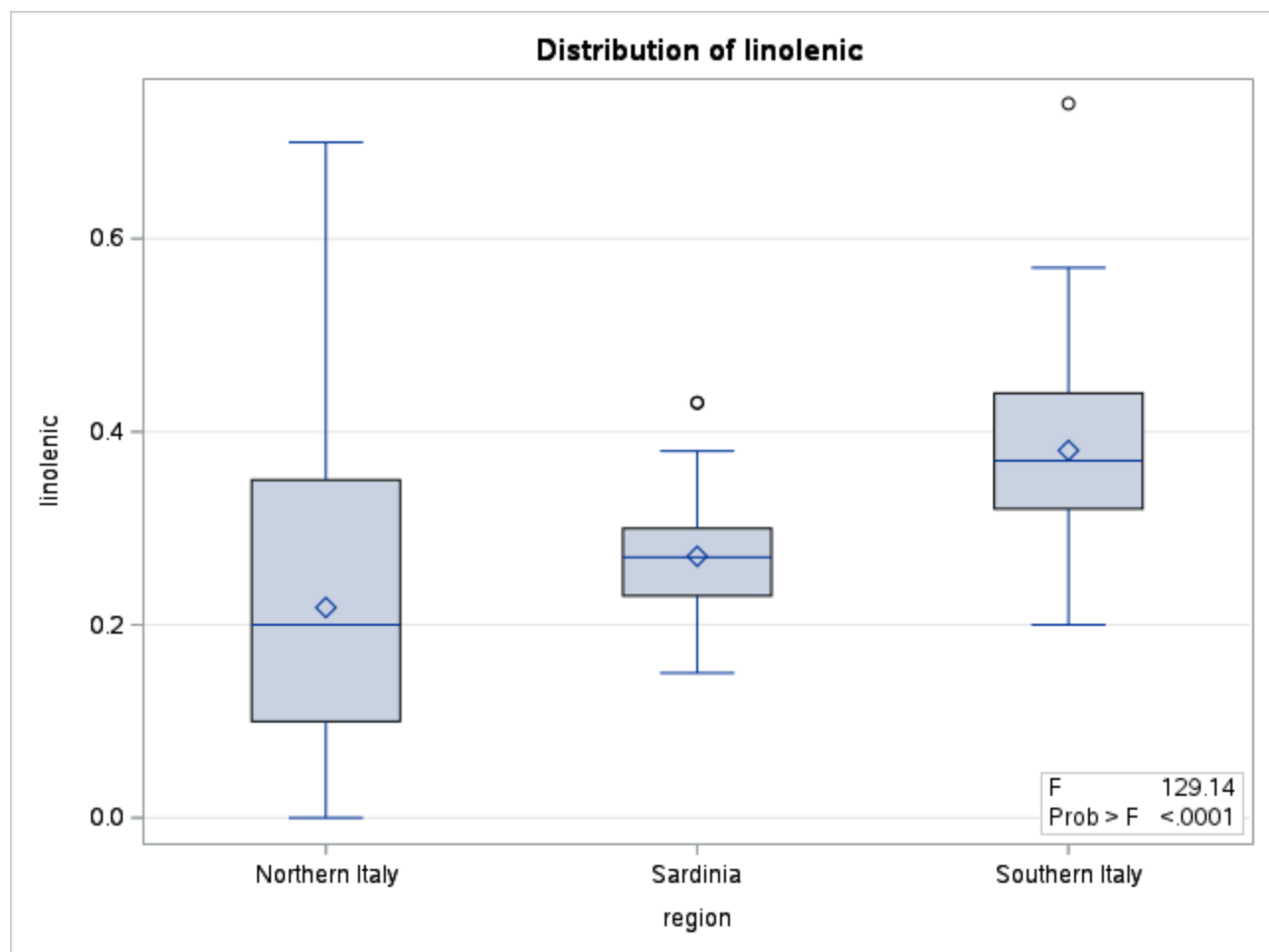
Dependent Variable: linolenic

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	2.99818125	1.49909062	129.14	<.0001
Error	569	6.60530267	0.01160862		
Corrected Total	571	9.60348392			

R-Square	Coeff Var	Root MSE	linolenic Mean
0.312197	33.78792	0.107743	0.318881

Source	DF	Type I SS	Mean Square	F Value	Pr > F
region	2	2.99818125	1.49909062	129.14	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
region	2	2.99818125	1.49909062	129.14	<.0001



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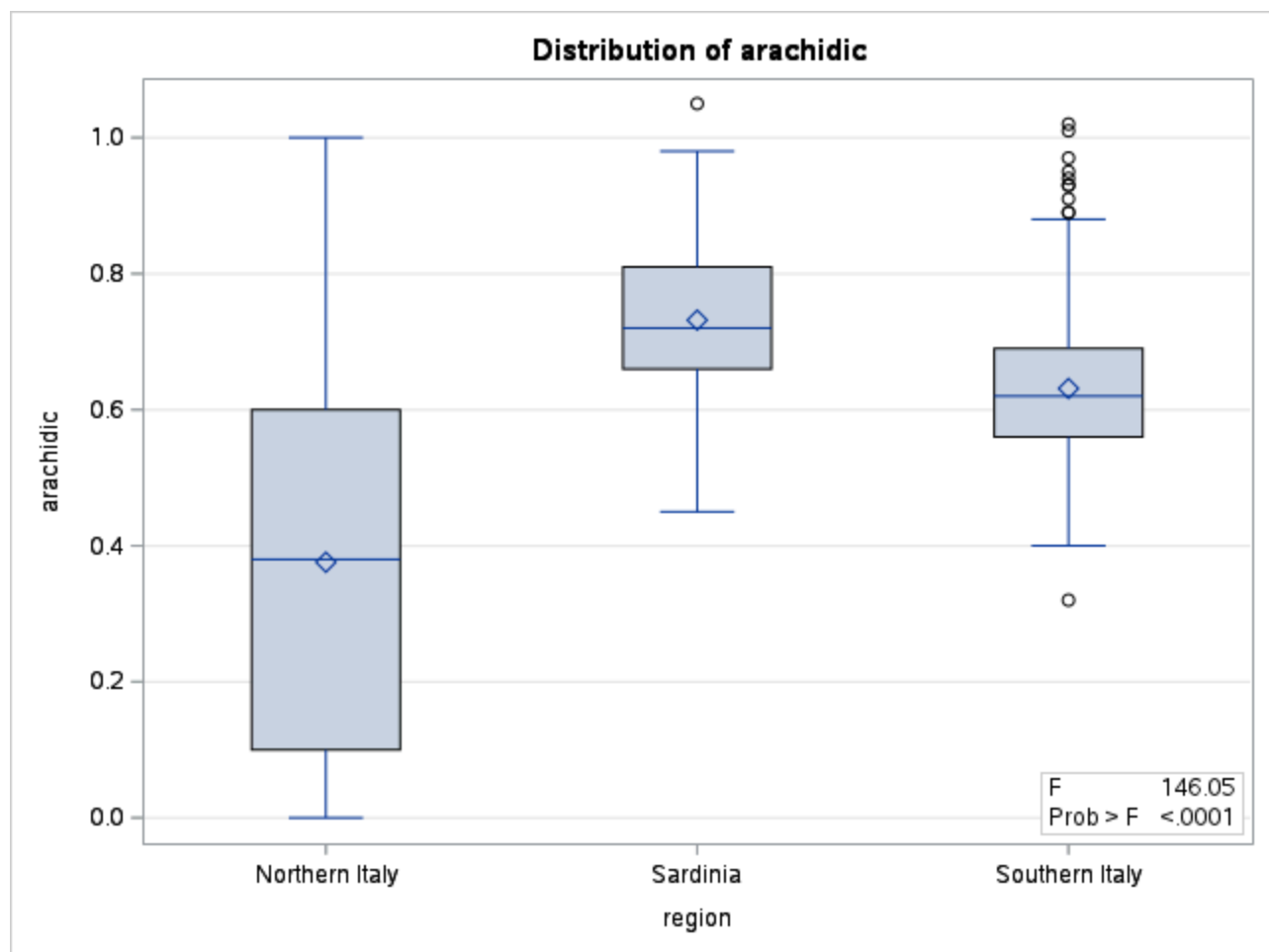
Dependent Variable: arachidic

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	9.40040629	4.70020314	146.05	<.0001
Error	569	18.31204546	0.03218286		
Corrected Total	571	27.71245175			

R-Square	Coeff Var	Root MSE	arachidic Mean
0.339212	30.87819	0.179396	0.580979

Source	DF	Type I SS	Mean Square	F Value	Pr > F
region	2	9.40040629	4.70020314	146.05	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
region	2	9.40040629	4.70020314	146.05	<.0001



The GLM Procedure

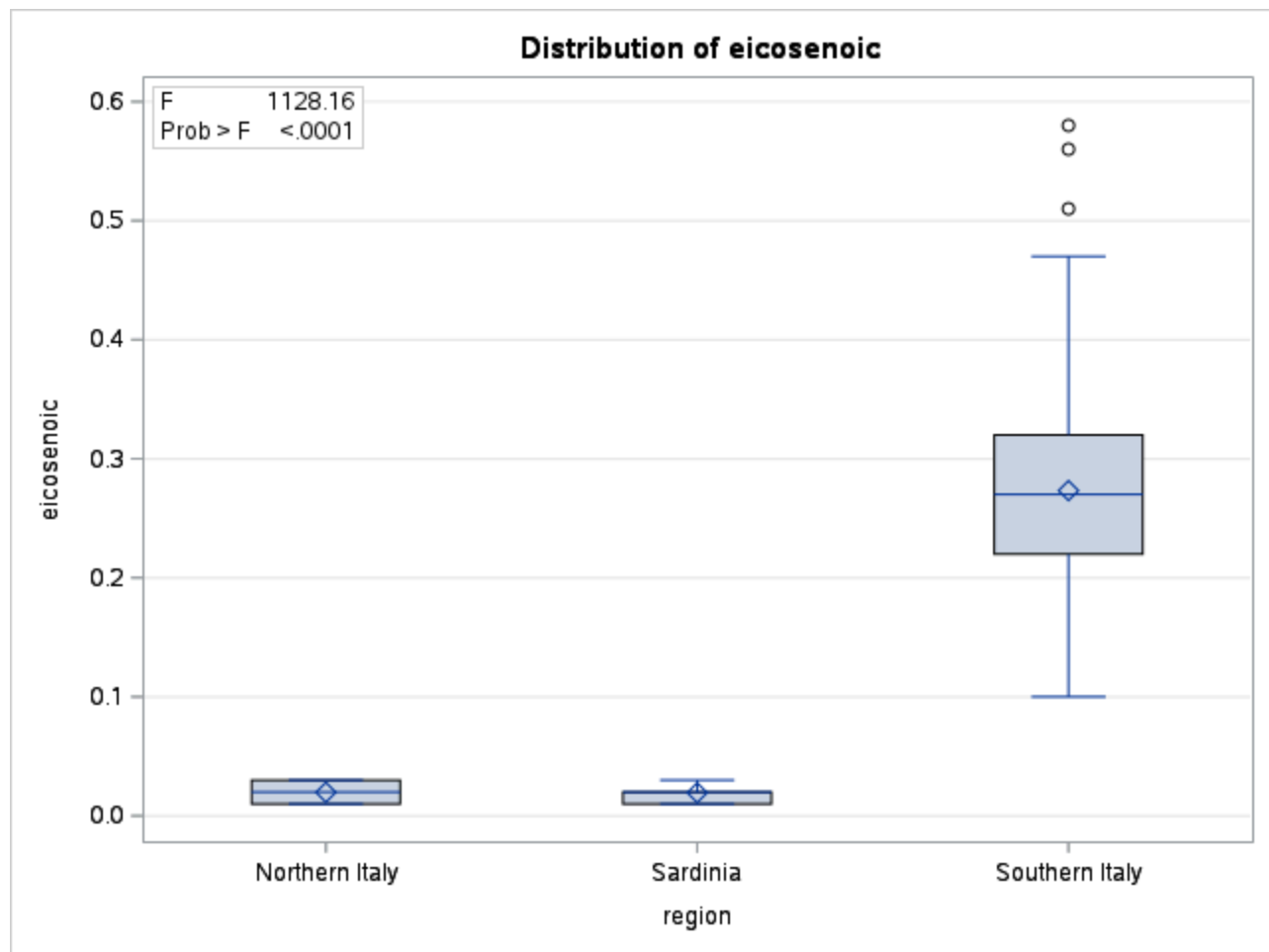
Dependent Variable: eicosenoic

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	9.04436429	4.52218215	1128.16	<.0001
Error	569	2.28080406	0.00400844		
Corrected Total	571	11.32516836			

R-Square	Coeff Var	Root MSE	eicosenoic Mean
0.798607	38.88609	0.063312	0.162815

Source	DF	Type I SS	Mean Square	F Value	Pr > F
region	2	9.04436429	4.52218215	1128.16	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
region	2	9.04436429	4.52218215	1128.16	<.0001



The GLM Procedure
Multivariate Analysis of Variance

E = Error SSCP Matrix								
	palmitic	palmitoleic	stearic	oleic	linoleic	linolenic	arachidic	eicosenoic
palmitic	871.21984667	206.81661537	-59.13671866	-1639.816397	635.42305965	-6.73795265	7.4669445633	-14.31226573
palmitoleic	206.81661537	95.193334531	-23.91988196	-545.2822776	303.27234907	-9.947801991	-7.80000875	-5.93869834
stearic	-59.13671866	-23.91988196	76.968523516	80.878990008	-87.52961706	0.7021735453	-0.883224945	4.2218811983
oleic	-1639.816397	-545.2822776	80.878990008	4438.5022507	-2486.801182	48.031431664	26.602869085	34.934036843
linoleic	635.42305965	303.27234907	-87.52961706	-2486.801182	1847.9381992	-49.40331834	-54.61738424	-25.91045229
linolenic	-6.73795265	-9.947801991	0.7021735453	48.031431664	-49.40331834	6.6053026687	6.6956405273	0.9721941728
arachidic	7.4669445633	-7.80000875	-0.883224945	26.602869085	-54.61738424	6.6956405273	18.31204546	1.7177110158
eicosenoic	-14.31226573	-5.93869834	4.2218811983	34.934036843	-25.91045229	0.9721941728	1.7177110158	2.2808040625

Partial Correlation Coefficients from the Error SSCP Matrix / Prob > r								
DF = 569	palmitic	palmitoleic	stearic	oleic	linoleic	linolenic	arachidic	eicosenoic
palmitic	1.000000	0.718155 <.0001	-0.228369 <.0001	-0.833898 <.0001	0.500790 <.0001	-0.088821 0.0340	0.059117 0.1587	-0.321071 <.0001
palmitoleic	0.718155 <.0001	1.000000	-0.279447 <.0001	-0.838880 <.0001	0.723079 <.0001	-0.396714 <.0001	-0.186820 <.0001	-0.403036 <.0001
stearic	-0.228369 <.0001	-0.279447 <.0001	1.000000	0.138376 0.0009	-0.232089 <.0001	0.031142 0.4581	-0.023526 0.5751	0.318644 <.0001
oleic	-0.833898 <.0001	-0.838880 <.0001	0.138376 0.0009	1.000000	-0.868319 <.0001	0.280518 <.0001	0.093313 0.0259	0.347205 <.0001
linoleic	0.500790 <.0001	0.723079 <.0001	-0.232089 <.0001	-0.868319 <.0001	1.000000	-0.447163 <.0001	-0.296906 <.0001	-0.399105 <.0001

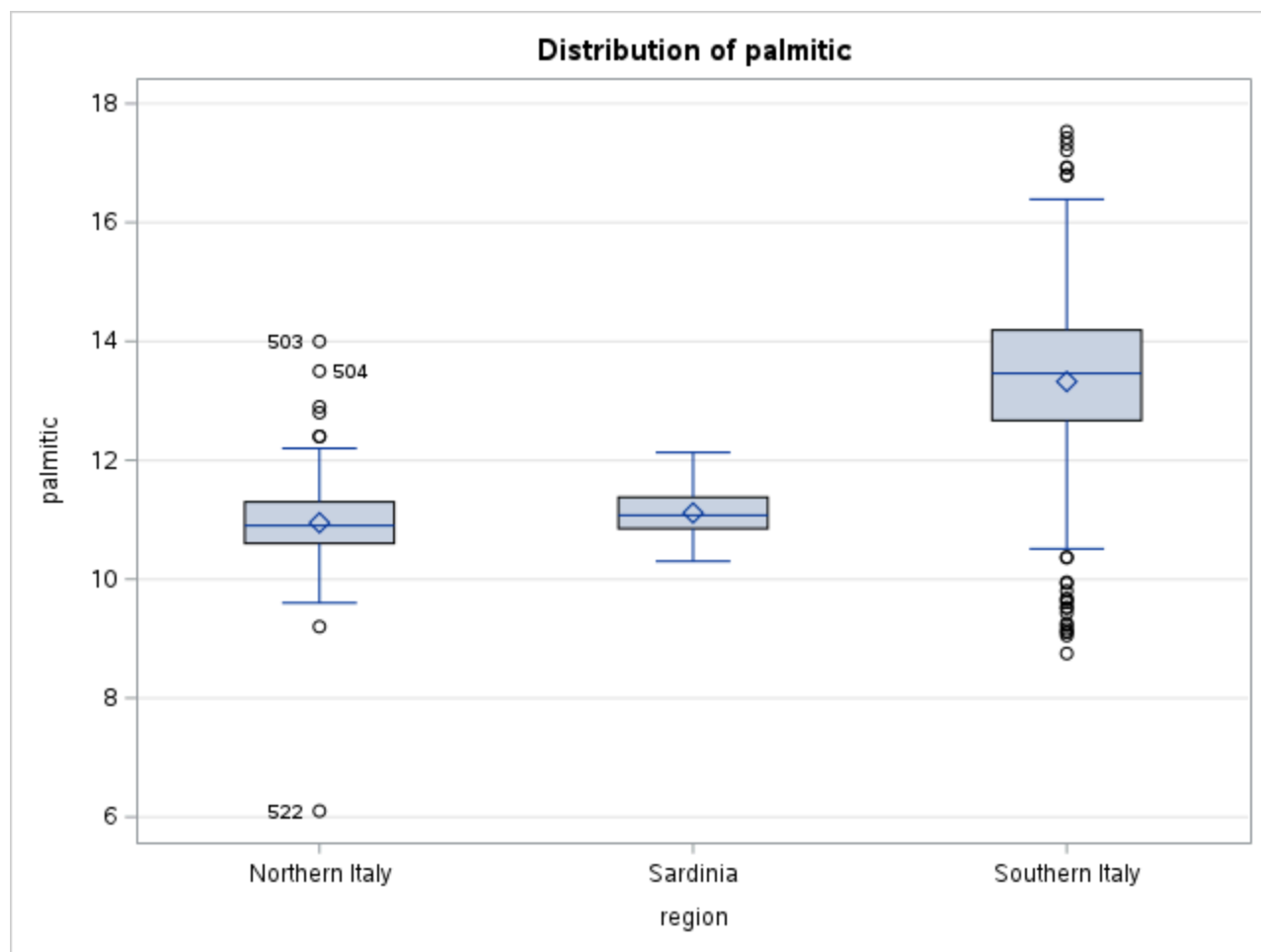
Partial Correlation Coefficients from the Error SSCP Matrix / Prob > r								
DF = 569	palmitic	palmitoleic	stearic	oleic	linoleic	linolenic	arachidic	eicosenoic
linolenic	-0.088821 0.0340	-0.396714 <.0001	0.031142 0.4581	0.280518 <.0001	-0.447163 <.0001	1.000000	0.608804 <.0001	0.250474 <.0001
arachidic	0.059117 0.1587	-0.186820 <.0001	-0.023526 0.5751	0.093313 0.0259	-0.296906 <.0001	0.608804 <.0001	1.000000	0.265789 <.0001
eicosenoic	-0.321071 <.0001	-0.403036 <.0001	0.318644 <.0001	0.347205 <.0001	-0.399105 <.0001	0.250474 <.0001	0.265789 <.0001	1.000000

The GLM Procedure
Multivariate Analysis of Variance

H = Type III SSCP Matrix for region								
	palmitic	palmitoleic	stearic	oleic	linoleic	linolenic	arachidic	eicosenoic
palmitic	751.75352396	215.45068183	-1.135873649	-1631.301229	441.35105433	46.604196706	40.950004388	82.364131468
palmitoleic	215.45068183	62.154755679	-0.551691119	-491.6126629	149.13089988	13.567306187	13.444680078	23.51437841
stearic	-1.135873649	-0.551691119	0.1273399455	15.844071531	-13.24341755	-0.187435084	-1.010921209	-0.073913891
oleic	-1631.301229	-491.6126629	15.844071531	4964.8363241	-2297.163215	-113.5935429	-189.9371593	-173.3475837
linoleic	441.35105433	149.13089988	-13.24341755	-2297.163215	1518.1902561	39.076097361	119.05342759	43.296351939
linolenic	46.604196706	13.567306187	-0.187435084	-113.5935429	39.076097361	2.9981812473	3.4226860461	5.0590072258
arachidic	40.950004388	13.444680078	-1.010921209	-189.9371593	119.05342759	3.4226860461	9.4004062878	4.1048127604
eicosenoic	82.364131468	23.51437841	-0.073913891	-173.3475837	43.296351939	5.0590072258	4.1048127604	9.0443642942

Characteristic Roots and Vectors of: E Inverse * H, where H = Type III SSCP Matrix for region E = Error SSCP Matrix									
Characteristic Root	Percent	Characteristic Vector V'EV=1							
		palmitic	palmitoleic	stearic	oleic	linoleic	linolenic	arachidic	eicosenoic
8.49408559	78.53	0.01146748	0.05394312	-0.01183751	0.00230442	0.00446112	0.17230915	-0.07374628	0.68422164
2.32246144	21.47	0.03761833	0.07604354	0.01784217	0.02628855	-0.00512713	0.02327395	-0.14636300	0.04340598
0.00000000	0.00	0.09358835	0.12414018	0.11620227	0.10068131	0.10080553	0.53925493	0.00126418	-0.02916145
0.00000000	0.00	-0.00067526	0.02054810	0.12043256	0.00133487	0.00169487	0.00159571	0.00549175	-0.03220323
0.00000000	0.00	-0.03460670	0.15241596	0.00506268	0.00337647	0.00127923	0.00481707	-0.00677260	-0.02210142
0.00000000	0.00	0.08654410	0.12868637	0.09251746	0.08137018	0.06451730	-0.10883082	0.29023576	0.05792572
0.00000000	0.00	0.10243188	0.03439752	0.08653468	0.10321516	0.10672790	-0.07723316	0.07768876	0.45374583
0.00000000	0.00	0.10426820	0.01801221	0.08729186	0.07093197	0.07600713	0.00000000	0.00000000	0.00000000

MANOVA Tests for the Hypothesis of No Overall region Effect H = Type III SSCP Matrix for region E = Error SSCP Matrix		
S=2 M=2.5 N=280		
Statistic	Value	P-Value
Wilks' Lambda	0.03170202	<.0001
Pillai's Trace	1.59368960	<.0001
Hotelling-Lawley Trace	10.81654703	<.0001
Roy's Greatest Root	8.49408559	<.0001



The GLM Procedure

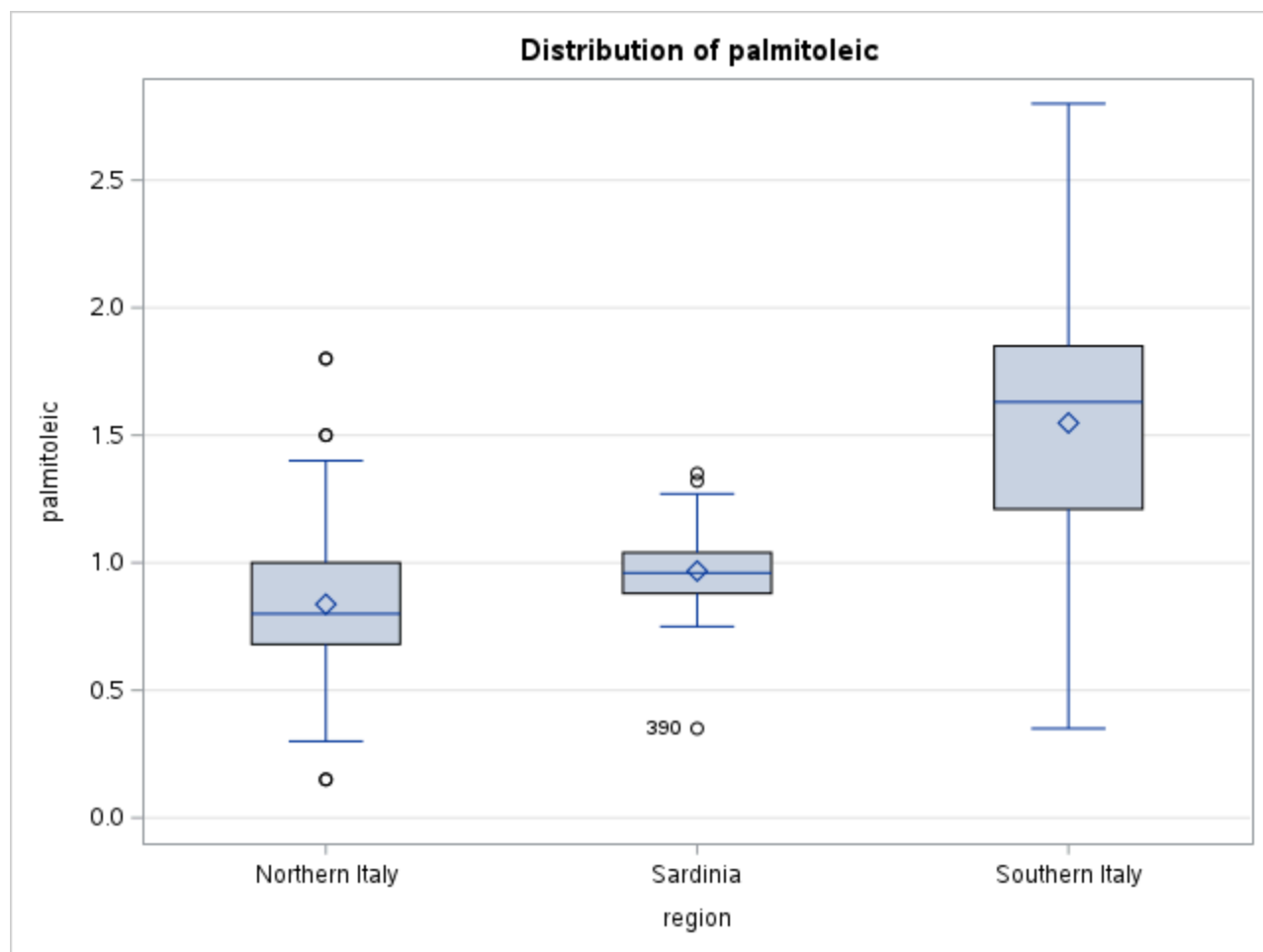
Tukey's Studentized Range (HSD) Test for palmitic

Note: This test controls the Type I experimentwise error rate.

Alpha	0.05
Error Degrees of Freedom	569
Error Mean Square	1.531142
Critical Value of Studentized Range	3.32320

Comparisons significant at the 0.05 level are indicated by ***.				
region Comparison	Difference Between Means	Simultaneous 95% Confidence Limits		
Southern Italy - Sardinia	2.20941	1.87408	2.54474	***
Southern Italy - Northern Italy	2.37487	2.08822	2.66151	***
Sardinia - Southern Italy	-2.20941	-2.54474	-1.87408	***
Sardinia - Northern Italy	0.16546	-0.21172	0.54264	
Northern Italy - Southern Italy	-2.37487	-2.66151	-2.08822	***
Northern Italy - Sardinia	-0.16546	-0.54264	0.21172	

The GLM Procedure



The GLM Procedure

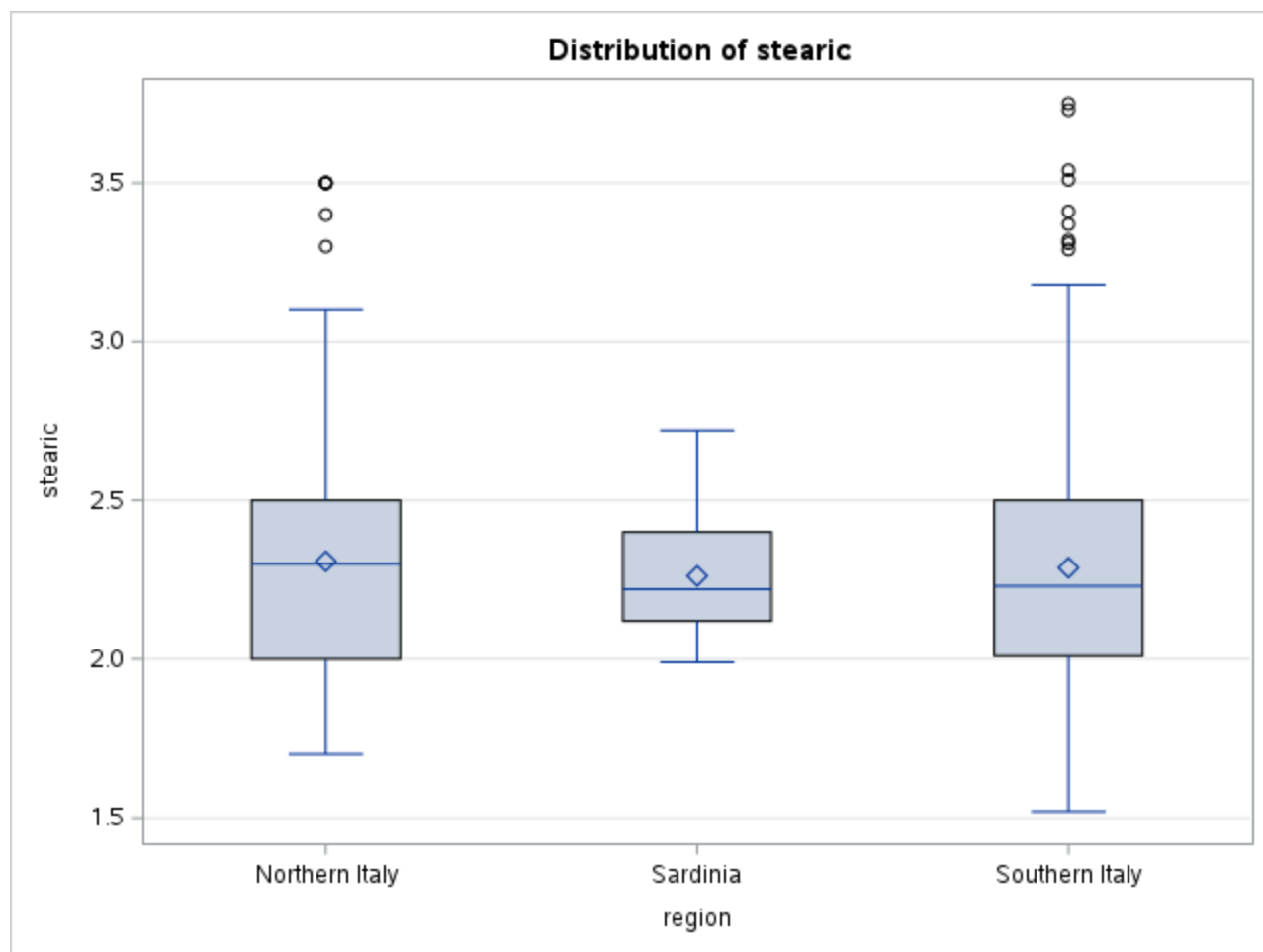
Tukey's Studentized Range (HSD) Test for palmitoleic

Note: This test controls the Type I experimentwise error rate.

Alpha	0.05
Error Degrees of Freedom	569
Error Mean Square	0.167299
Critical Value of Studentized Range	3.32320

Comparisons significant at the 0.05 level are indicated by ***.				
region Comparison	Difference Between Means	Simultaneous 95% Confidence Limits		
Southern Italy - Sardinia	0.58057	0.46972	0.69141	***
Southern Italy - Northern Italy	0.71067	0.61592	0.80542	***
Sardinia - Southern Italy	-0.58057	-0.69141	-0.46972	***
Sardinia - Northern Italy	0.13010	0.00542	0.25478	***
Northern Italy - Southern Italy	-0.71067	-0.80542	-0.61592	***
Northern Italy - Sardinia	-0.13010	-0.25478	-0.00542	***

The GLM Procedure



The GLM Procedure

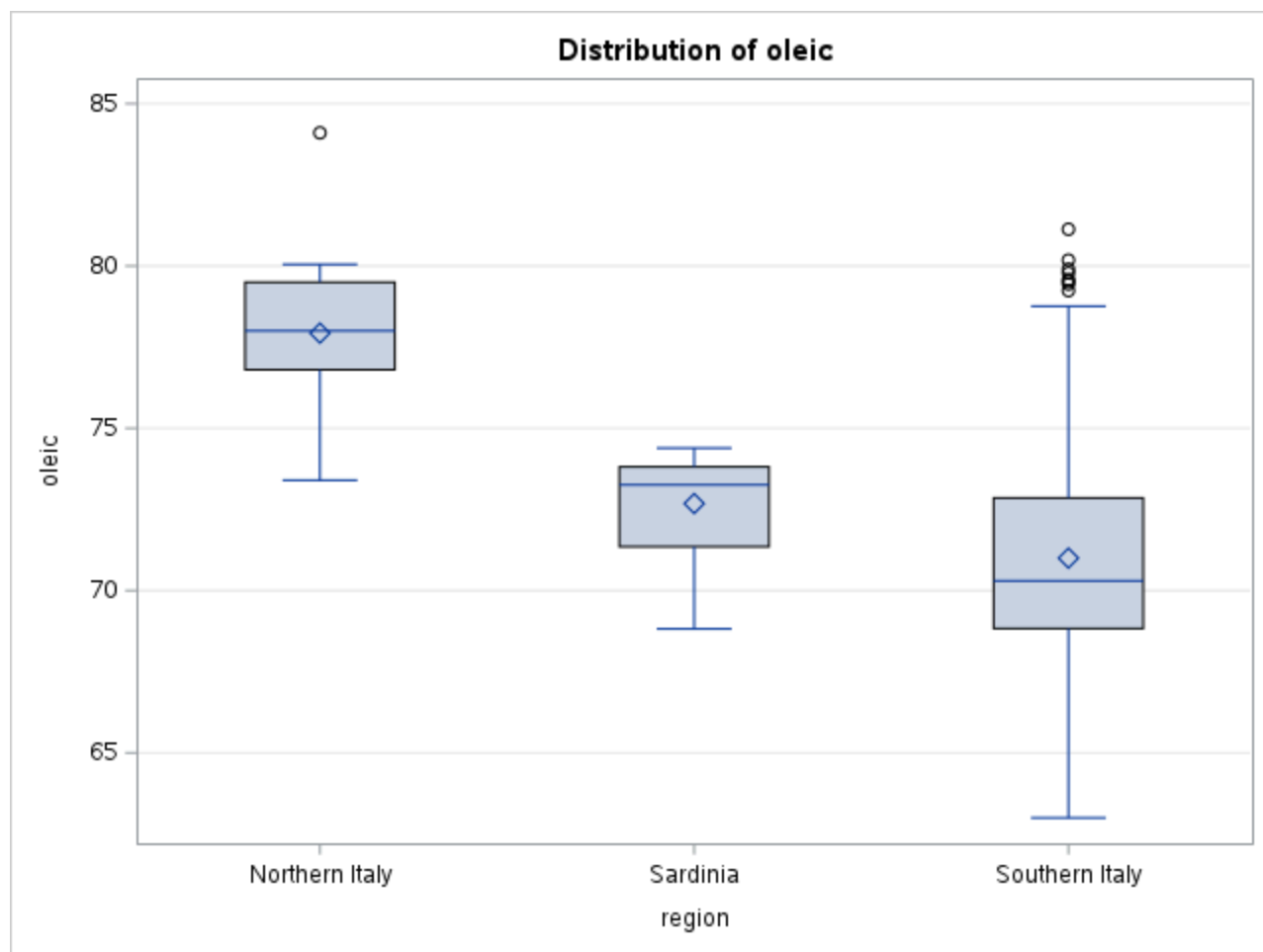
Tukey's Studentized Range (HSD) Test for stearic

Note: This test controls the Type I experimentwise error rate.

Alpha	0.05
Error Degrees of Freedom	569
Error Mean Square	0.13527
Critical Value of Studentized Range	3.32320

Comparisons significant at the 0.05 level are indicated by ***.				
region Comparison	Difference Between Means	Simultaneous 95% Confidence Limits		
Northern Italy - Southern Italy	0.02027	-0.06493	0.10547	
Northern Italy - Sardinia	0.04618	-0.06593	0.15829	
Southern Italy - Northern Italy	-0.02027	-0.10547	0.06493	
Southern Italy - Sardinia	0.02590	-0.07377	0.12557	
Sardinia - Northern Italy	-0.04618	-0.15829	0.06593	
Sardinia - Southern Italy	-0.02590	-0.12557	0.07377	

The GLM Procedure



The GLM Procedure

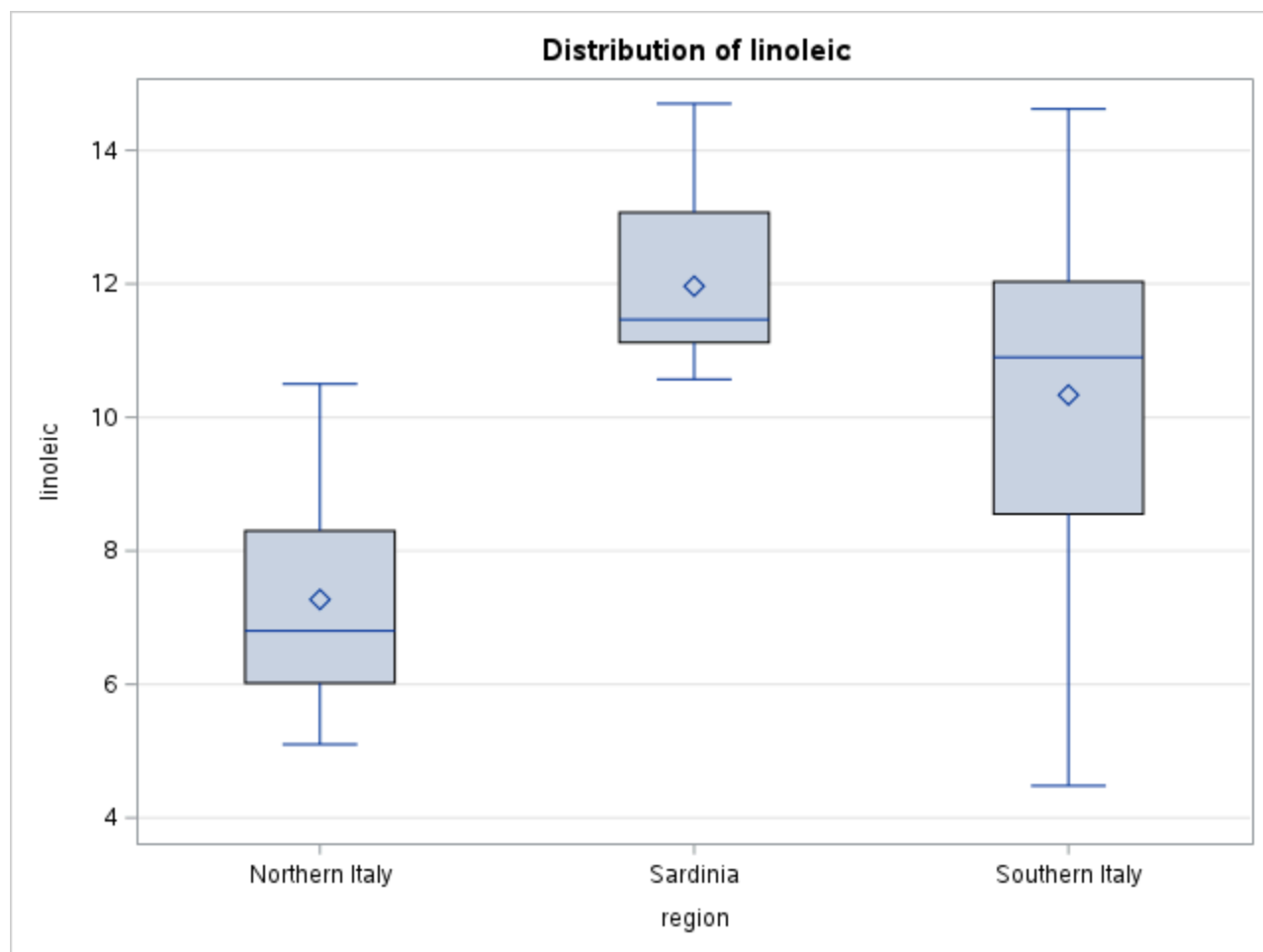
Tukey's Studentized Range (HSD) Test for oleic

Note: This test controls the Type I experimentwise error rate.

Alpha	0.05
Error Degrees of Freedom	569
Error Mean Square	7.800531
Critical Value of Studentized Range	3.32320

Comparisons significant at the 0.05 level are indicated by ***.				
region Comparison	Difference Between Means	Simultaneous 95% Confidence Limits		
Northern Italy - Sardinia	5.2503	4.3990	6.1017	***
Northern Italy - Southern Italy	6.9304	6.2834	7.5774	***
Sardinia - Northern Italy	-5.2503	-6.1017	-4.3990	***
Sardinia - Southern Italy	1.6801	0.9232	2.4370	***
Southern Italy - Northern Italy	-6.9304	-7.5774	-6.2834	***
Southern Italy - Sardinia	-1.6801	-2.4370	-0.9232	***

The GLM Procedure



The GLM Procedure

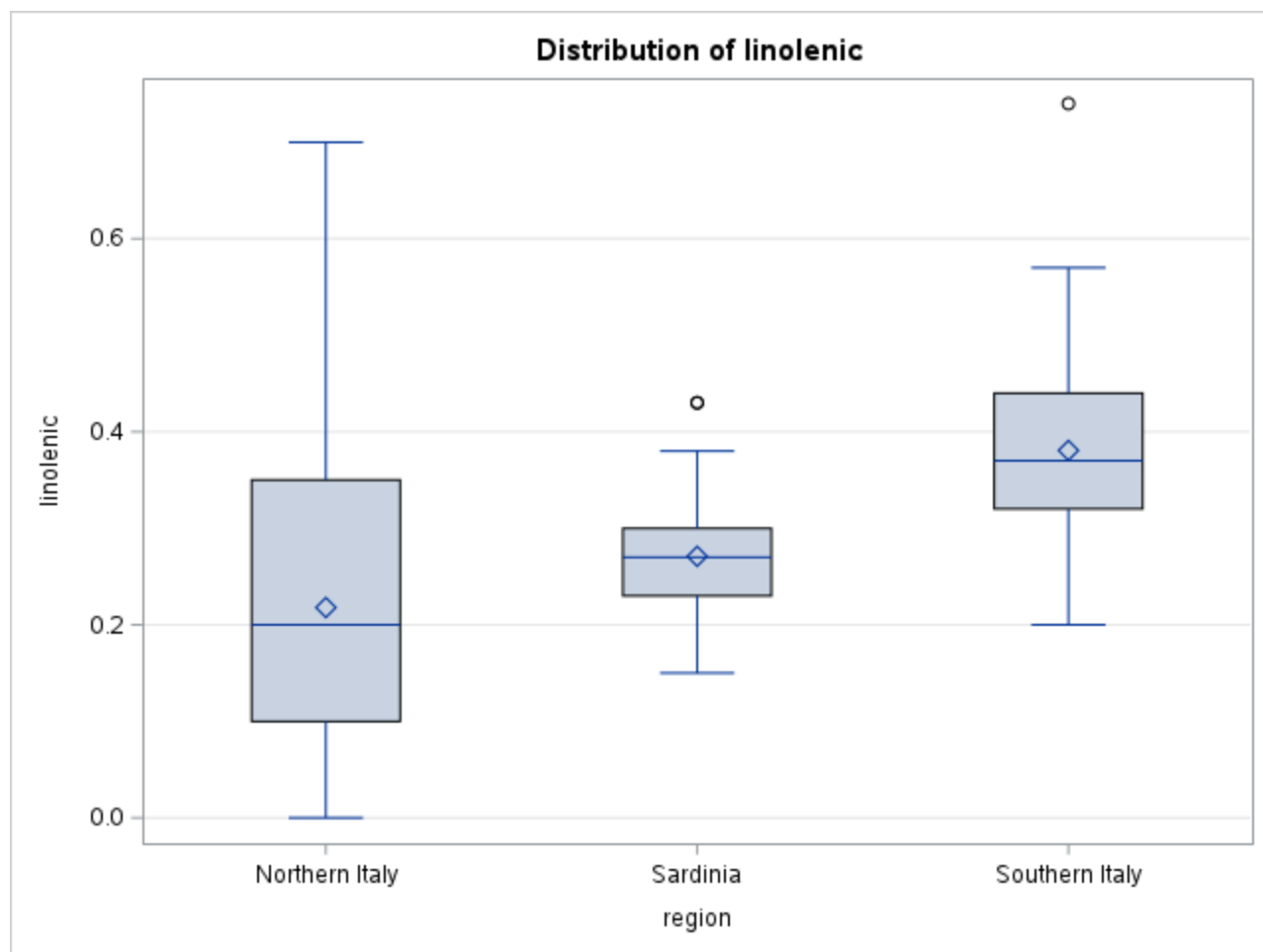
Tukey's Studentized Range (HSD) Test for linoleic

Note: This test controls the Type I experimentwise error rate.

Alpha	0.05
Error Degrees of Freedom	569
Error Mean Square	3.247695
Critical Value of Studentized Range	3.32320

Comparisons significant at the 0.05 level are indicated by ***.				
region Comparison	Difference Between Means	Simultaneous 95% Confidence Limits		
Sardinia - Southern Italy	1.6303	1.1419	2.1187	***
Sardinia - Northern Italy	4.6950	4.1457	5.2443	***
Southern Italy - Sardinia	-1.6303	-2.1187	-1.1419	***
Southern Italy - Northern Italy	3.0647	2.6472	3.4821	***
Northern Italy - Sardinia	-4.6950	-5.2443	-4.1457	***
Northern Italy - Southern Italy	-3.0647	-3.4821	-2.6472	***

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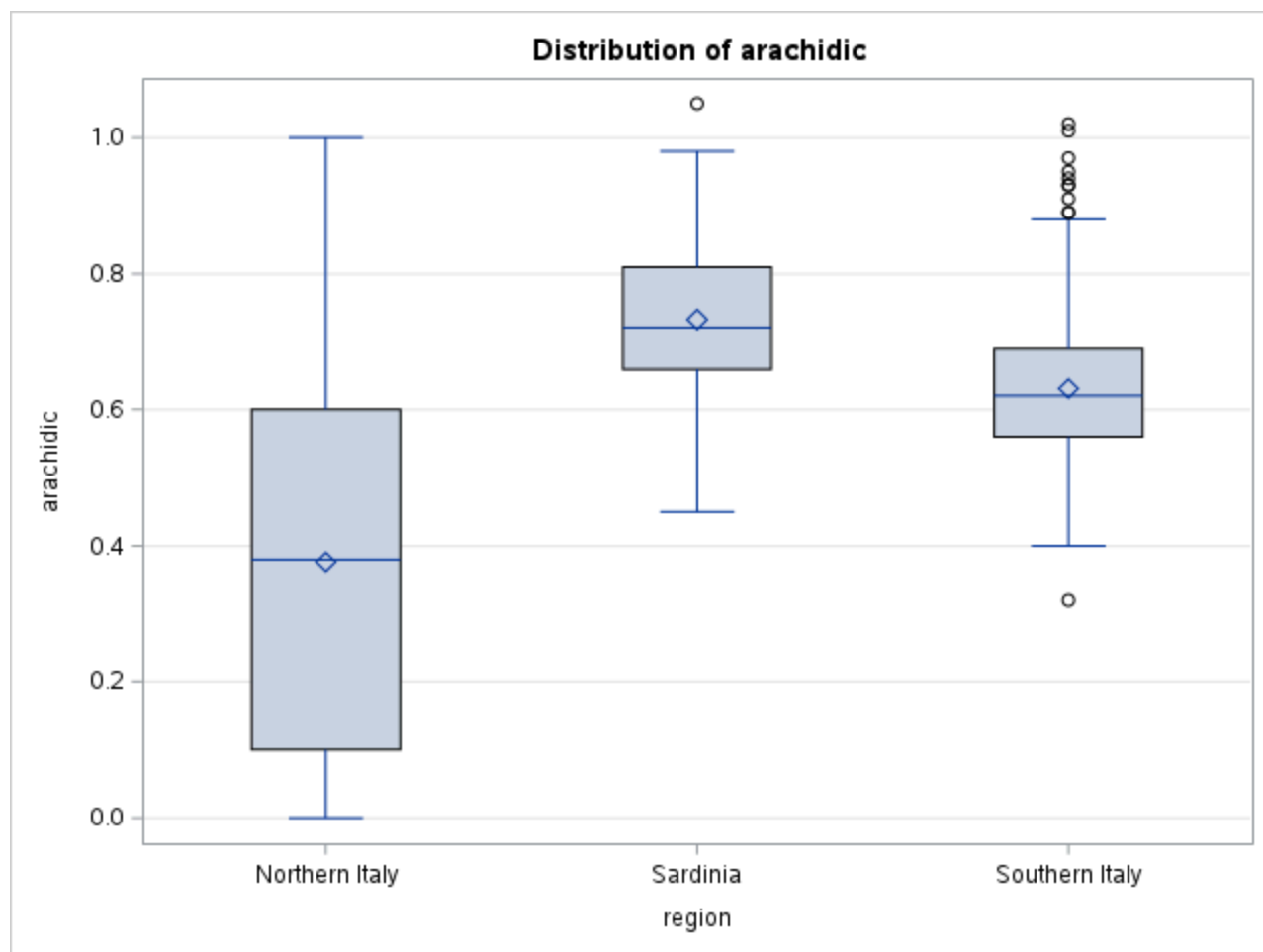
Tukey's Studentized Range (HSD) Test for linolenic

Note: This test controls the Type I experimentwise error rate.

Alpha	0.05
Error Degrees of Freedom	569
Error Mean Square	0.011609
Critical Value of Studentized Range	3.32320

Comparisons significant at the 0.05 level are indicated by ***.				
region Comparison	Difference Between Means	Simultaneous 95% Confidence Limits		
Southern Italy - Sardinia	0.109732	0.080533	0.138930	***
Southern Italy - Northern Italy	0.162769	0.137810	0.187729	***
Sardinia - Southern Italy	-0.109732	-0.138930	-0.080533	***
Sardinia - Northern Italy	0.053038	0.020196	0.085880	***
Northern Italy - Southern Italy	-0.162769	-0.187729	-0.137810	***
Northern Italy - Sardinia	-0.053038	-0.085880	-0.020196	***

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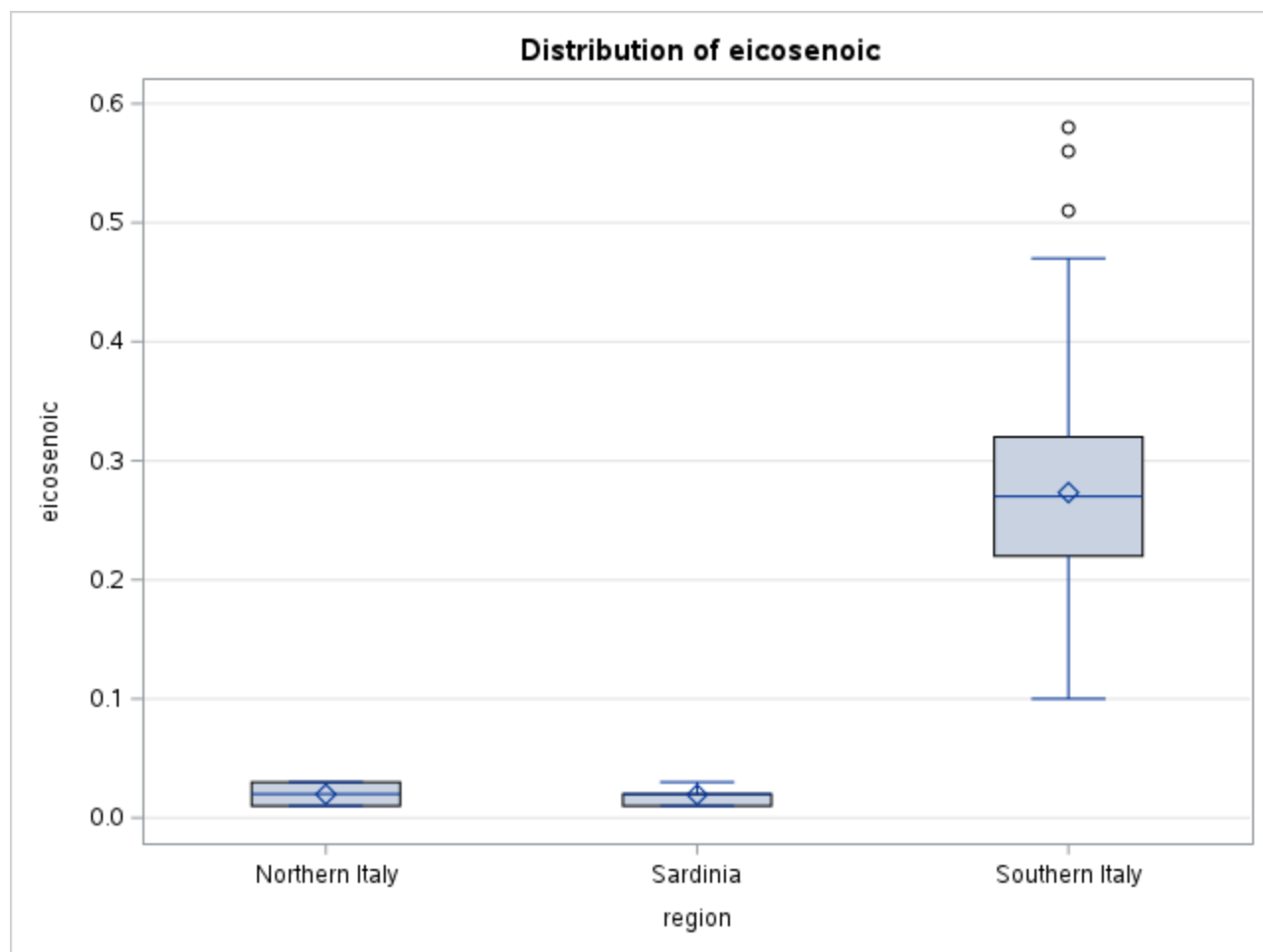
Tukey's Studentized Range (HSD) Test for arachidic

Note: This test controls the Type I experimentwise error rate.

Alpha	0.05
Error Degrees of Freedom	569
Error Mean Square	0.032183
Critical Value of Studentized Range	3.32320

Comparisons significant at the 0.05 level are indicated by ***.				
region Comparison	Difference Between Means	Simultaneous 95% Confidence Limits		
Sardinia - Southern Italy	0.10056	0.05194	0.14917	***
Sardinia - Northern Italy	0.35597	0.30129	0.41066	***
Southern Italy - Sardinia	-0.10056	-0.14917	-0.05194	***
Southern Italy - Northern Italy	0.25541	0.21386	0.29697	***
Northern Italy - Sardinia	-0.35597	-0.41066	-0.30129	***
Northern Italy - Southern Italy	-0.25541	-0.29697	-0.21386	***

The GLM Procedure



The GLM Procedure

Tukey's Studentized Range (HSD) Test for eicosenoic

Note: This test controls the Type I experimentwise error rate.

Alpha	0.05
Error Degrees of Freedom	569
Error Mean Square	0.004008
Critical Value of Studentized Range	3.32320

Comparisons significant at the 0.05 level are indicated by ***.				
region Comparison	Difference Between Means	Simultaneous 95% Confidence Limits		
Southern Italy - Northern Italy	0.253485	0.238818	0.268151	***
Southern Italy - Sardinia	0.253832	0.236674	0.270990	***
Northern Italy - Southern Italy	-0.253485	-0.268151	-0.238818	***
Northern Italy - Sardinia	0.000347	-0.018951	0.019646	
Sardinia - Southern Italy	-0.253832	-0.270990	-0.236674	***
Sardinia - Northern Italy	-0.000347	-0.019646	0.018951	